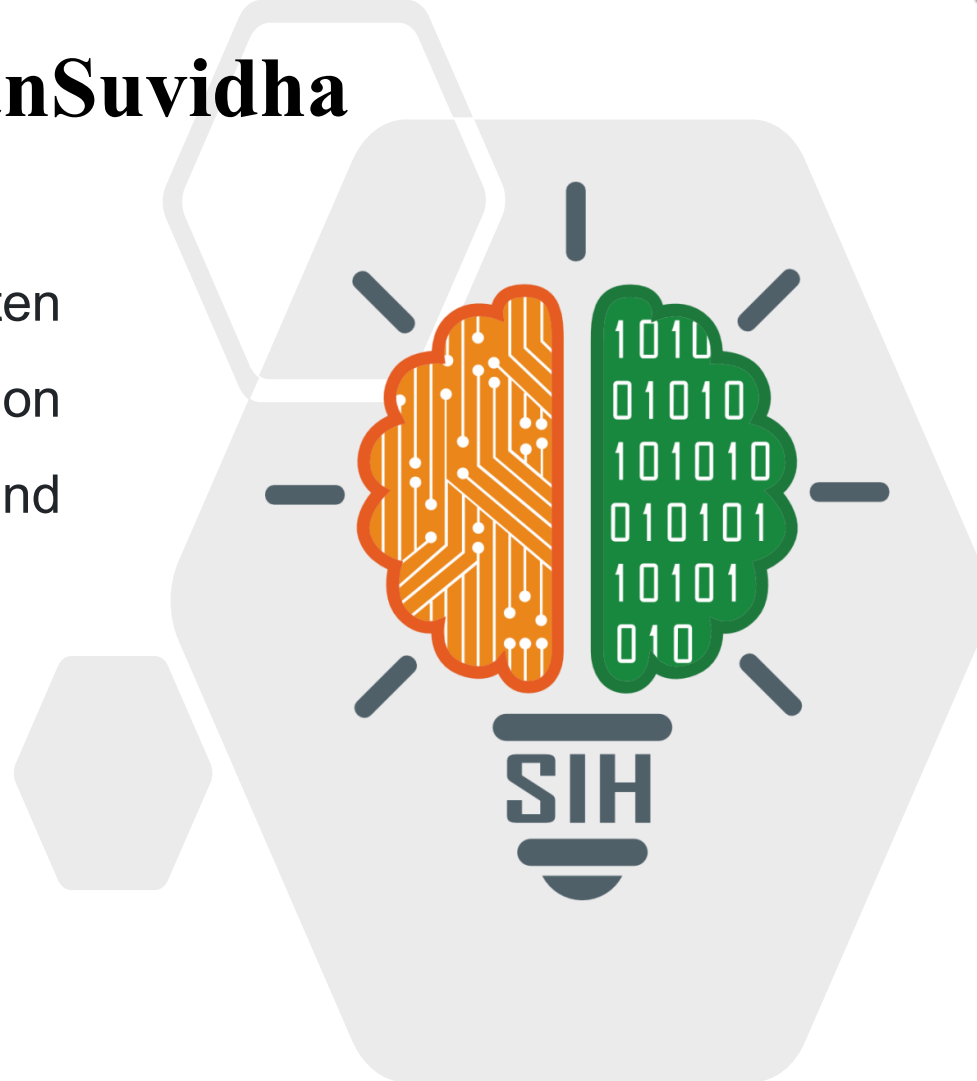


SMART INDIA HACKATHON 2026



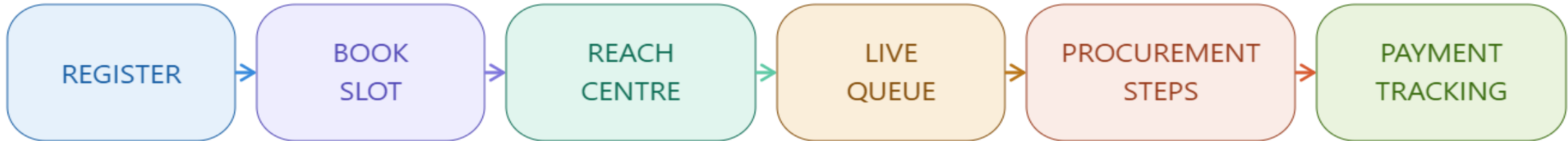
Title: KisanSuvidha

- **Problem Statement ID** – 26032
- **Problem Statement Title** - Farmers often face long waiting times, lack of information regarding procurement schedules, and uncertainty about procurement status.
- **Theme** - Smart Automation
- **PS Category** - Software
- **Team ID** -
- **Team Name** - Byte Brigade



Your Crop Deserves Less Waiting.

A digital procurement platform enabling farmers to book slots, track live queues, receive timely alerts, and monitor procurement and payment status—reducing congestion and waiting time at procurement centres.



Problems Facing:

- Long & unpredictable queues at procurement centres.
- No real-time information about queue status and centre capacity.
- **Overcrowding** causes stress, extra costs and **possible crop damage**.
- **Uncertainty** about procurement and payment status.

Our Proposed Solution: **KisanSuvidha**

- Digital slot booking & farmer registration for planned visits.
- **Live queue tracking** with estimated waiting time.
- **Nearest procurement centre finder** with center availability.
- **SMS/App Notifications** and procurement & payment status tracking.

Target Users:

Farmers (Primary user), Centre Admin & District Official

TECHNICAL APPROACH

KisanSuvidha Procurement Core

TECH STACK

Frontend



HTML



CSS



React

Backend



Node.js

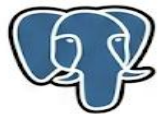


JavaScript



Express

DataBase



PostgreSQL

Notifications & Messaging



MSG91



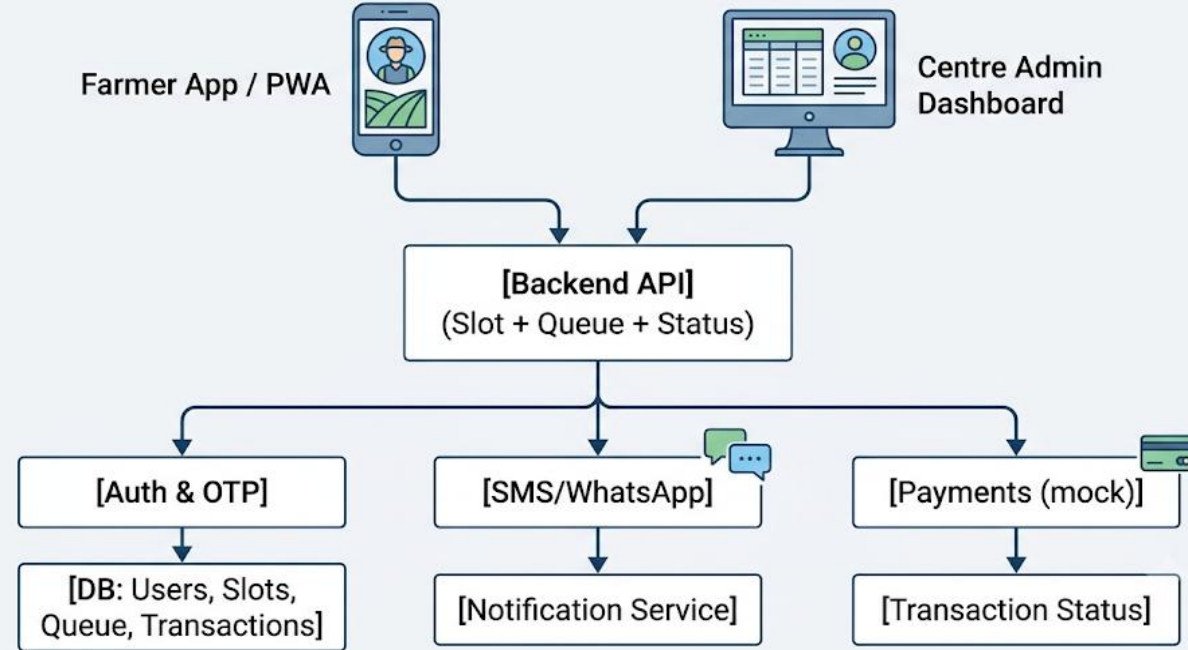
SMS



WhatsApp
Business
Cloud API
WhatsApp (Meta)

Farmer App / PWA

Centre Admin
Dashboard



Real Time & Location:

- **Socket.IO:** Tracks Queue Status
- **Leaflet Map API:** Nearest Procurement Centres

Security & Access:

- **JWT/OTP:** Secure Login Authentication
- **Role Based Access:** Farmer, Center Admin, District Official

FEASIBILITY AND VIABILITY

KisanSuvidha Integrated Feasibility and Viability Framework

1. Technical Feasibility



2. Operational Feasibility



3. Financial Viability



4. Market & Policy Viability



1. Technical Feasibility:

- **Standard Technology:** APIs, databases, notifications and real-time queue systems.
- **Rural Friendly:** Simple UI, local languages, offline support, SMS/IVR.

2. Operational Feasibility:

- **Easy Staff Management:** Simple dashboard for slots, tokens and status.
- **Farmer Convenience:** Less Waiting, clear slots and timely notifications.

3. Financial Viability:

- **Affordable Deployment:** Low development, hosting and maintenance costs.
- **Sustainable Model:** Government/FPO contracts, grants or CSR funding.

4. Market & Policy Viability:

- **Government aligned:** Support Digital India, e-NAM and AgriStack.
- **High Demand:** Useful for APMCS, FCI, NAFED, FPOs and farmers.

IMPACT AND BENEFITS



Farmer Empowerment

- **Less Waiting and Better Planning:** Pre-booked slots **reduces waiting**.
- **Transparency & Financial Security:** **Reduces dependency** on middlemen.

Streamlined Procurement

- **Better Operations:** Staggered slots **reduce crowding** and help centres managing staffs & resources.
- **Transparency & Accountability:** Digital records reduce disputes, favouritism and **corruption risks**.



Metric	Current (Typical)	Target with KisanSuvidha
Avg waiting time at mandi	6 to 8 hours	1 to 2 hours
Farmer trips to check status	2 to 3 visits	0 (all digital)
Payment delay complaints	High	Reduced by 60% to 70%
Centre Staff overtime	Frequent during peaks	Reduced by 30% to 40%
Farmer Satisfaction	Low to Medium	High

RESEARCH AND REFERENCES

Sources / References	Derived Insights for our Projects
Central Food Grains Procurement Portal (CFPP)- FCI: https://www.cfpp.nic.in/#/KMS/2025-2026/4/2	The national portal tracks farmer registration, procurement, MSP payments, procurement centres and payment timelines , showing the need for centralized procurement monitoring .
Department of Food & Public Distribution – Procurement Policy: https://dfpd.gov.in/procurement-policy/en	Wheat and paddy are procured through FCI and State agencies at MSP , with procurement based on production, marketable surplus and crop patterns. This supports our centre-wise capacity and scheduling approach .
e-Uparjan – Government of Madhya Pradesh: https://www.india.gov.in/services	The system sends SMS alerts about the data and place to bring grain and manages procurement related payment. This supports our notification system .
e-NAM – National Agriculture Market: https://enam.gov.in/	e-NAM digitizes mandi trading for transparent pricing and faster payments, supporting our registration, procurement tracking, and payment-status features .
Digital Agriculture Mission: Tech for Transforming Farmers' Lives: https://static.pib.gov.in/WriteReadData/specificdocs/documents/2024/sep/doc202494386501.pdf	Supports our idea of using digital infrastructure to provide farmers with faster, more transparent, farmer-centric services.